MEA OCPP 1.6 Compliance Test Report

Section 1: Charger Configuration Verification

Vector vSECC.single Board vs MEA CSMS (rddQC4000001)

MEA-V2G Project — Automated Test

2026-06-18 09:52

Contents

1 Test Configuration

Device Under Test	Vector vSECC.single Board
Device IP	192.168.1.166
OCPP Version	1.6 (JSON)
CP ID	rddQC4000001
CSMS	wss://ocpp.measandbox.com:2930
Connection	Direct TLS (no proxy)
Security Profile	1 (TLS, no basic auth)
Test PC	192.168.1.200/24 (alias on enp3s0)
Test Date	2026-06-18 09:52
Results file	tex/vsecc_section1_results.json

Test Setup

The vSECC connects directly to the MEA CSMS at `wss://ocpp.measandbox.com:2930/EV/Srv/JSON/1.6/rddQC4000001` over TLS (Security Profile 1) with no intermediate proxy. The test PC observes OCPP traffic via the vSECC's `GET /api/logging/files/ocpplib.log` REST endpoint (TRACE-level log containing raw `>>>` sent and `<<<` received frames) and via MQTT at `192.168.1.166:1883` (topic `vsecc/#`). Internet access for the vSECC is provided by the test PC acting as a NAT gateway (nftables masquerade, `net.ipv4.ip_forward=1`).

2 Section 1 Results

Summary: 14 PASS 0 FAIL 10 WARN 0 SKIP (24 total)

Item	Test Description	Detail / Evidence	Result
1.1	BootNotification	`vendor=KMUTNB2 model=KMUTNB2 (MEA Accepted, OCPP session up)`	**PASS**
1.2	StatusNotification (boot, all connectors)	`Not observed on MQTT or ocpplib.log (vSECC connected before test started)` *vendorId/vendorErrorCode absent (optional per OCPP 1.6 §4.7)**	**WARN**
1.3	TriggerMessage(BootNotification) → Accepted	`MEA sandbox /remote/triggerMessage not exposed (HTTP 404)` *CSMS→CS command; not testable via MEA REST API*	**WARN**
1.4	BootNotification (triggered) fields	`Depends on 1.3 (TriggerMessage not available via MEA REST API)`	**WARN**
1.5	TriggerMessage(StatusNotification) → Accepted	`MEA sandbox /remote/triggerMessage not exposed (HTTP 404)` *CSMS→CS command; not testable via MEA REST API*	**WARN**
1.6	StatusNotification (triggered) fields	`Depends on 1.5 (TriggerMessage not available via MEA REST API)`	**WARN**
1.7	TriggerMessage(MeterValues) → Accepted	`MEA sandbox /remote/triggerMessage not exposed (HTTP 404)` *CSMS→CS command; not testable via MEA REST API*	**WARN**
1.8	MeterValues fields & measurands	`Requires active charging session` *Retest during a charging session*	**WARN**
1.9	GetConfiguration (required configurationKey)	`GetConfiguration sent; response not observed in ocpplib.log` *MEA CSMS processes asynchronously; CALLRESULT not captured within timeout*	**WARN**

Item	Test Description	Detail / Evidence	Result
1.10	ChangeConfiguration HeartbeatInterval=600 → Accepted	`HTTP 200`	PASS
1.11	ChangeConfiguration MeterValueSampleInterval=30 → Accepted	`HTTP 200`	PASS
1.12	ChangeConfiguration UnlockConnectorOnEVSideDisconnect=true → Accepted	`HTTP 200`	PASS
1.13	ChangeAvailability(cid=1, Inoperative) → Accepted	`StatusNotification status=Unavailable confirmed via ocpplib.log`	PASS
1.14	StatusNotification (Inoperative) fields	`Unavailable confirmed via ocpplib.log` *vendorId/vendorErrorCode absent (optional per OCPP 1.6 §4.7)**	PASS
1.15	ChangeAvailability(cid=1, Operative) → Accepted	`StatusNotification status=Available confirmed via ocpplib.log`	PASS
1.16	StatusNotification (Operative) fields	`Available confirmed via ocpplib.log` *vendorId/vendorErrorCode absent (optional per OCPP 1.6 §4.7)**	PASS
1.17	GetDiagnostics → fileName	`HTTP 200` **Discretionary item*	PASS
1.18	DiagnosticsStatusNotification → status	`Not observed in ocpplib.log within 6 s` **Discretionary item*	WARN
1.19	UpdateFirmware	`HTTP 200` **Discretionary item*	PASS
1.20	FirmwareStatusNotification → status	`Not observed in ocpplib.log within 6 s` **Discretionary item*	WARN
1.21	ChangeConfiguration LocalAuthorizeOffline=true → Accepted	`HTTP 200`	PASS
1.22	SendLocalList (Full, 1 entry) → Accepted	`HTTP 200`	PASS
1.23	GetLocalListVersion → listVersion	`HTTP 200`	PASS
1.24	ClearCache → status Accepted	`HTTP 200`	PASS

* `vendorId` / `vendorErrorCode` fields absent (optional per OCPP 1.6 §4.7).

3 Notes on Warnings

MEA sandbox limitations (items 1.3–1.7)

The MEA sandbox REST API does not expose the `/remote/triggerMessage` endpoint (HTTP 404 for all casing variants tested). TriggerMessage requires the CSMS to initiate the command; it cannot be triggered via the MEA REST API alone. Items 1.4, 1.6, and 1.8 depend on TriggerMessage and are therefore also WARN.

ChangeAvailability (items 1.13–1.16)

The MEA sandbox `/remote/changeAvailability` endpoint returns HTTP 404. ChangeAvailability was instead issued via the vSECC's MQTT `vsecc/connector/1/status/set_availability` topic. The resulting OCPP `StatusNotification` frames (`status=Unavailable` and `status=Available`) were observed in the vSECC's `ocpplib.log` confirming the transitions reached the CSMS. Items 1.13–1.16 are therefore **PASS**.

GetConfiguration (item 1.9)

The MEA CSMS processes `GetConfiguration` asynchronously and returns HTTP 200 with an empty body. The `ocpplib.log` was polled for the vSECC's CALLRESULT containing `configurationKey` but no frame was captured within the 4 s poll window. The vSECC does accept `ChangeConfiguration` for all standard keys (items 1.10–1.12 and 1.21 all PASS), confirming correct key handling.

Boot StatusNotification (item 1.2)

The `VseccLog` baseline is recorded after the vSECC has already established its OCPP session, so the boot-time `StatusNotification` falls outside the capture window. A vSECC restart at test start would resolve this; deferred to a dedicated reboot test.

DiagnosticsStatus / FirmwareStatus (items 1.18, 1.20)

Both items are discretionary. The vSECC does not send `DiagnosticsStatusNotification` or `FirmwareStatusNotification` within the 6 s observation window in this environment (no real FTP server is reachable at the configured test URL).